

Name..... Centre /Index No

School Signature

P530/3
BIOLOGY
(Practical)
PAPER 3
July/August 2025
3¹/₄ hours



WAKISSHA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

BIOLOGY PRACTICAL

Paper 3

3 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

- *This paper consists of **three** questions.*
- *Answer **all** questions.*
- *Answers must be written in the spaces provided.*
- *Additional sheets of paper must **not** be inserted in this booklet.*

FOR EXAMINERS' USE ONLY		
Question	Marks	Examiner's Initials & No.
1		
2		
3		
Total		

Question 1

75 MINUTES (40 marks)

- (a) You are provided with specimen V.

Place the specimen provided on the dissection board with the dorsal side up and hind side nearest to you, Extend the right hind limb using a pin. Dissect the right hind limb from pelvic region to the ankle, symmetrically, remove the skin to expose the underlying structures.

- (i) Describe the adaptations of the exposed structures to suit the different functions.

(07½ marks)

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- (ii) Draw and label the exposed structures.

(09 ½ marks)

- (b) Now turn the specimen with the ventral side uppermost, stretch out the legs and pin them on the dissection board. Ligature the abdominal vein and continue to dissect to expose the internal structures after stretching out the skin.

Displace

- (i) The alimentary canal to the left of the specimen.
- (ii) The heart anteriorly and leave the kidney in the undisturbed state.

Draw and label blood vessels draining;

- Head
- Thoracic and abdominal region
- And the right hind limb

(23 marks)

Question 2**60 MINUTES (30 marks)**

You are provided with solution X, Y and D and organism Q. Carry out tests to establish the identity of the nature of solutions. (13½marks)

Table A

Test	Solution	Observation	Deductions
To 1cm ³ of solution Add 3 drops of Iodine	X		
	Y		
To 1cm ³ of solution Add 1cm ³ of Benedicts solutions and boil	X		
	Y		

Now label three test tubes 1,2,3 and treat them as follows.

Test tube 1 add 1cm³ of solution X + a level spatula of Q

Test tube 2 add 1cm³ of solution Y + a level spatula of Q

Test tube 3 add 1cm³ of solution Y + spatula of Q + 1cm³ of D

Shake the test tube to mix thoroughly.

Put the test tube in a water bath maintained at 35 – 40°C for 30 minutes. Subject the test tubes to the following test and record your observations in table B below;

Table B

Test	Test tube	Observations	Deductions
Starch test	1		
	2		
	3		
Reducing sugar test	1		
	2		
	3		

Now mark test tubes 4 and 5 and treat them as follows;

Test tube 4 add solution X + level spatula of Q and boil, cool the mixture.

Test tube 5 add solution Y + level spatula of Q + 1 cm³ of D + 1 cm³ of Hcl

Put these test tubes in a water bath maintained at 35 – 40°C for 30 minutes and then treat them as below. Record your observations and deductions in table C below.

Table C

Test	Test tube	Observations	Deductions
Starch test	4		
	5		
Reducing sugar test	4		
	5		

(a) Explain your deductions in the table B for the test tubes.

- 1..... (2½ marks)
- 2..... (2½ marks)
- 3..... (2½ marks)

(b) Compare tables A, B and C suggest explanations for results in

Test tubes 1 and 4 (3 marks)

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Test tubes 3 and 5 (3 marks)

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(c) From the experiments in the table suggest the favourable conditions under which organisms Q works best. (3 marks)

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Question 3**60 MINUTES (30 marks)**

You are provided with coloured beads representing individuals of different phenotype in a population. White beads represent homozygous individuals (WW) for white allele. Red represents homozygous individuals red alleles (RR) and pink beads represent heterozygotes. (WR).

Procedure:

- Place 10 white beads, 10 red beads and 20 pink beads in the envelope provided. Shake well to mix thoroughly, without looking there choose 30 of these beads at random, drawing one at a time and shaking the envelope thoroughly before each pick. Let this be $n = 30$
- Put back the beads in the envelope shake thoroughly to mix and draw 20 beads again at random, while shaking the envelope contents before a draw. (Remember to draw one bead at a time). Let this be $n = 20$
- Repeat the above steps, this time drawing 10 beads. Let this be $n = 10$

Record your observations table D below

(07 marks)

Beads picked	White beads	Pick beads	Red beads
$n = 30$			
$n = 20$			
$n = 10$			

Using the formula;

$$\text{allele frequency for white (p)} = \frac{2 \times \text{no. of white beads picked} + \text{no. of heterozygotes (pink)}}{\text{total number of individuals in a population.}}$$

$$\text{allele frequency for red (q)} = \frac{2 \times \text{no. of red beads picked} + \text{no. of heterozygotes (pink)}}{\text{total number of individuals in a population.}}$$

Calculate the allele frequencies in each pick (n) and fill the table E below.

(6 marks)

Table E

Expected frequency	Actual frequencies (number of beads drawn)		
	$n = 30$	$n = 20$	$n = 10$
$P = 0.5$			
$Q = 0.5$			

- (a) (i) What happened to the actual frequencies when the number of beads drawn is decreased. (2 marks)

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- (ii) Assuming the draws represent only individuals to reproduce in the generation. Explain the effect of reduced reproductive individuals in the population basing on the changes in the alleles, in the table E above, with relation to the expected frequency. (4 marks)

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- (b) (i) A phenomenon used to describe what is happening in a(ii) above where there is reduced reproductive potential in a population due to chance (drawing randomly) alone is called. (1 mark)

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- (ii) What type of inheritance is being depicted by the pink individuals? Explain (3 marks)

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- (c) If a grey bead replaces one of the white beads, representing a mutation in one of the parental white allele. Establish the genotype corresponding to new colours, grey, dark red and black that would be introduced. (3 marks)

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- (d) With reference to the background in the environment in which the population of individuals live / exist. How could conditions change to favour the rare black allele?

(4 marks)

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END

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PRACTICAL INSTRUCTIONS

CONFIDENTIAL:

This information is given only to facilitate preparation of the examination.

Great care should be taken that the information given below does not reach the candidates either directly or indirectly.

INSTRUCTIONS FOR SPECIMENS AND APPARATUS.

The teacher responsible for preparing specimens and apparatus must ensure that candidates are provided with correct specimens and other materials as specified in these instructions.

Specimens and solutions which have been assigned codes should be presented to candidates using those **codes only** and not any other identity.

The head teacher **must** ensure that the teacher responsible for preparing the specimens and apparatus **hands in his/her trial results for physiology/ biochemistry question properly sealed in a separate envelope and firmly fastened (attached) to the candidates' scripts envelope(s).**

Each candidate must be provided with;

V – Freshly killed Frog/toad

Solutions X – 10% glucose solution

Y – 1% starch solution

Q – Yeast (fresh)

D – 1% amylase solution

Provide Candidates with

Test-tube

10 large white beads

20 large pink beads

10 large red beads

} Should be of same size

- Test tubes
- Food test reagents
- Heat source
- Water bath
- Dissecting board
- Envelope (Medium sized)

END